

The Nature of Infinitesimal Measurement: Measuring with Surreal Numbers

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Abstract

In this paper we introduce a transfinite extension of the notion of physical measurement. Our construction mirrors the Peano-Jordan measure but crucially makes use of the Dedekind-cut simplicity-hierarchy structure available in the class **No.** of surreal numbers. An operation on subclasses of **No.** stratified by the infinite ordinals $\alpha \in \mathbf{On.}$ which we call the α -Estimate is defined, to be thought of as the "best" estimate for the size of the set or class on day α . A short introduction to the theory and construction of the surreal number system is given, followed by the main definitions for the α -Estimate. We then show that our definition has many desirable properties and perform some computations. We then extend the α -Estimate to higher (finite) dimensions. Finally we compare our estimate operator $[\]_\alpha$ to the Lebesgue measure on the reals μ , in particular we show that we can approximate any finite approximations of the Borel-like classes, but that certain **No.** analogues of Borel sets will never agree with μ no matter how large α becomes. We conclude with interesting further directions for research.

1 A Short Introduction to surreal Numbers

1.1 History

The surreal number system, denoted **No.** named by Donald Knuth [11] is an extension of the real numbers $\mathbb{R}$, discovered by John H. Conway [6] as a fragment of a larger study of perfect information, two-player games [7]. **No.** contains the familiar class of ordinal numbers $\omega, \omega+1, \dots, 2*\omega, \dots, \alpha, \dots$, as well as more exotic numbers such as $\omega - 1, \ln(\omega - \pi), \frac{1}{e*\omega}$. It comes equipped with extensions of familiar operations $+, -, \times, \div$ with all of the typical properties one would expect. Most interesting is the method of constructing the class of surreals, we will primarily follow [6], highlighting features of the construction which will be of particular import to us in this paper.

1.2 Construction of the surreals

Definition 1.1. A surreal number is a pair of sets of surreal numbers L, R , where every element of the *left set* L is strictly less than every element of the *right set* R . The pair is typically denoted $\{L \mid R\}$.

Upon first reading the definition seems circular, *how does the business get started?* is a common response. Note that the definition above begins with *sets* of surreal numbers, in particular, by vacuous truth the set $\emptyset$ is a set of surreal numbers, and each element of the left set is less than each element of the right set (whatever "less than" means), again, vacuously. Taking our initial pair $L = R = \emptyset$, we can start with the number

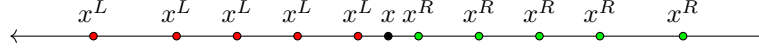
$$\{\emptyset \mid \emptyset\} = \{ \mid \} = 0$$

Note that this is the *definition* of 0, we could have just as well called this number anything that suited our fancy, but it would be an insult to any other number to define it this way. Note that we now have one new set of surreal numbers, namely $\{0\}$, and so two new numbers can be born, these are:

$$\{0 \mid \emptyset\} = 1 \text{ and } \{\emptyset \mid 0\} = -1$$

These numbers are well defined, again thanks to vacuous truth. Note once more that these are *definitions* and not strictly speaking correspondences to any previously defined number system. We will see later that these labels are justified once we have introduced some arithmetic operations.

For the following definition, if $x = \{L \mid R\}$ let x^L denote a typical member of L and x^R denote a typical member of R . One may think of the *left set* as a set of numbers which is less than x and the *right set* as a set of numbers greater than x , and so much like the Dedekind-cut construction of the reals, the pair L and R pick out of a value between the two sets. The following definition illustrates this intuition:

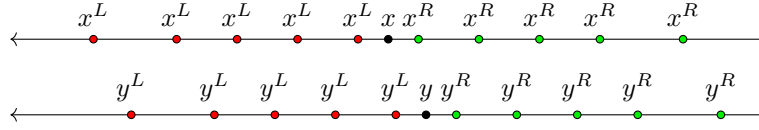


Definition 1.2. We say that a surreal $x \leq y$ if and only if no $y^R \leq x$ and $y \leq$ no x^L .

That is, loosely, $x \leq y$ if nothing greater than y is less than or equal to x and nothing smaller than x is greater than or equal to y . For those who have yet to encounter the surreals, it is a good exercise to prove that $x \leq x$ by our definition.

Definition 1.3. We say that $x = y$ if and only if $x \leq y$ and $y \leq x$.

Definition 1.4. We say that $x < y$ if and only if $x \leq y$ and $x \neq y$.



Definition 1.5. For two classes of numbers A, B say $A < B$ iff $\forall a \in A, \forall b \in B, a < b$.

Note that from the diagram above it is clear that $x \leq y$ because there is no y^R less than or equal to x and there is no x^L such that y is less than x^L . $y \not\leq x$ however, because there is an $x^R \leq y$. Alternatively, one could also cite that there is a y^L such that $x \leq y^L$. Now that the $<$ relation has been defined, we can conclude that $\{ 0 \mid 0 \}$ is not a number, because $\neg(0 < 0)$. Now that we have some new numbers, we can form new sets of numbers. The number

$$\{0, 1 \mid \} = 2$$

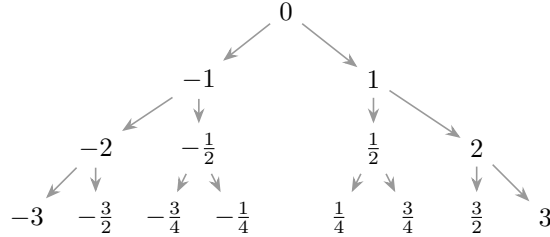
Can be defined, (exercise: prove that it is equal to the number $\{ 1 \mid \}$). We can also define

$$\{ 0 \mid 1 \} = \frac{1}{2}$$

Along with their corresponding negative counterparts $\{ \mid -1, 0 \} = -2$ and $\{ -1 \mid 0 \} = -\frac{1}{2}$.

Thus far, we have proceeded to construct numbers in stages. At stage 0 the number 0 was created, at stage 1 we saw -1 and 1 . At stage 2 we saw $-2, -\frac{1}{2}, \frac{1}{2}, 2$, at stage 3 we will see the numbers $-3, -1\frac{1}{2}, -\frac{3}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, 1\frac{1}{2}, 3$ (construct these!) At each stage we get new numbers that are more "complicated" than at the last stage.

Consider the number $\{ \frac{-1}{2} \mid \frac{3}{2} \}$. This number is **not** equal to $\frac{1}{2}$! You cannot just take the average!



Rather, $\{ \frac{-1}{2} \mid \frac{3}{2} \} = 0$ (prove this!) In general, a surreal number of the form $\{ L \mid R \}$ will evaluate to the value x , where x is the *simplest* number which satisfies $x^L < x < x^R$.

Definition 1.6. We say that x has birthday α if x is produced at stage α for the first time in our construction. We write $\beta(x) = \alpha$.

As per the figure above, $\frac{1}{2}$ has birthday 2, $\frac{3}{4}$ has birthday 3. In fact, we can continue construction indefinitely up to stage ω , on which some interesting numbers are born:

$$\{ 1, 2, 3, \dots \mid \} = \omega$$

$$\{ \mid -1, -2, -3, \dots \} = -\omega$$

Alongside

$$\{ 0 \mid 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots \} = \epsilon$$

And

$$\{ 0, \frac{1}{4}, \frac{5}{16}, \dots \mid 1, \frac{1}{2}, \frac{3}{8}, \dots \} = \frac{1}{3}$$

Indeed, at the finite stages only the dyadic rationals are born, but at stage ω , all of the rationals, and the rest of the reals too, are created. For more details, see [6]. For an alternative construction of the surreals, as functions from ordinals **On.** to the set $\{0, 1\}$, see [3]. For a construction of the surreals as games, see Part 1 of [6], [7] volumes 1-2.

1.3 Arithmetic Operations on surreals

Definition 1.7. $x + y = \{ x^L + y, x + y^L \mid x^R + y, x + y^R \}$

Definition 1.8. $-x = \{ -x^R \mid -x^L \}$

Much like the construction above, these arithmetic definitions rely on recursion and have at bottom a reliance on vacuous truth. So that for instance we see that $-0 = 0$ and $x + 0 = x$ as consequences of these definitions. We conclude this section with the definition of multiplication.

Definition 1.9.

$$xy = \{ x^L y + xy^L - x^L y^L, x^R y + xy^R - x^R y^R \mid x^L y + xy^R - x^L y^R, x^R y + xy^L - x^R y^L \}$$

This definition is by no means obvious. We summarize the argument given in [6] but refer the reader there for a greater exposition. Starting with (for instance) $x > x^L \implies x - x^L > 0$ (and similarly $y - y^L > 0$) Therefore $(x - x^L)(y - y^L) > 0$ and so $xy > x^L y + xy^L - x^L y^L$ and so this term should belong to the left set. One may proceed to discover all the terms in this manner. The figures below also give some motivation for this definition.

1.4 Gaps

Definition 1.10. A *gap* is a pair of classes $\mathcal{L}$ and $\mathcal{R}$ such that $\mathcal{L} < \mathcal{R}$ and at least one of $\mathcal{L}, \mathcal{R}$ is a proper class.

Some examples of classes are:

$$\Omega = \{ \mathbf{On.} \mid \}$$

$$\infty = \{ 1, 2, 3, \dots \mid \chi \text{ such that } \chi > n \in \mathbb{N} \text{ and } \chi \text{ is surreal} \}$$

Gaps cannot be regarded as numbers because either the left or right options constitute a proper class and not a set. Nonetheless they will arise naturally in later sections.

2 Main Definitions and Motivations

2.1 Motivation

Transfinite extensions of natural procedures have been considered in the theory of computation, utility theory [5], in the study of super tasks [12] [1] [8]. In the field of computation, this line of inquiry has yielded the study of infinite time turing machines [10]. Other work on transfinite register machines, which in the finite case are equal in power to Turing machines, turn out to have slightly different properties when strengthened [4]. Might there be a natural way to extend the everyday practice of measurement into the transfinite realm? This paper is meant to answer that question in one direction. There are already developments using non-standard analysis, see [9], here we try to use the natural structure afforded to us by the surreal number system, which have been used to extend the realm of study of probability [8], decision theory [5] and even hyper-computation [1]. We provide a rough outline before the formal work begins. One imagines that each day, one is sent some number of rulers, the lengths of these rulers correspond to the positive values in the construction of **No.** On day 0 we haven't any rulers, and thus cannot perform any measurements. On day 1, we have a ruler of length 1. On day 2, we have a ruler of length 2 and length $\frac{1}{2}$, and so on and so forth. Then, given a set X on some given day α , we first attempt to get an underestimate of its size, the range of possible measurements we can make is called the *internal set* $\text{In}_\alpha(X)$ defined below. Similarly, we find the set of all possible overestimates $\text{Out}_\alpha(X)$. At each stage we use only *finitely many* rulers. The α -Estimate is then defined as the simplest number which is between the sets $\text{In}_\alpha(X)$ and $\text{Out}_\alpha(X)$. Thus it is in some sense "the best estimation for the size of the set X on day α ." Thanks to the structure of **No.**, we need not limit ourselves to just finite values of α , and so we have achieved an extension of a kind.

2.2 Definitions

Definition 2.1. Let $\mathcal{M}$ be the class of all non-empty open intervals $[a, b]$ where $a, b \in \mathbf{No.}$ and $a < b$.

Definition 2.2. If $[a, b] = C \in \mathcal{M}$ say the length of C , denoted $|C| = b - a$.

It is clear to see that $|C|$, though it may be infinitesimal will always be strictly greater than 0 (take special note that $a \neq b$ because $a < b$). Indeed this is a natural choice, for when conducting measurements, one does not use rulers of zero-length.

Definition 2.3. If $\mathcal{M} \ni C = [a, b]$ we define the interior of C , denoted $\mathbf{int}(C) = (a, b)$.

Definition 2.4. We define the numbers *made* up to day α , denoted

$$\mathbf{M}(\alpha) := \{x \in \mathbf{No.} \mid \beta(x) \leq \alpha\}$$

It is clear that if $\alpha \leq \beta$ then $M(\alpha) \subseteq M(\beta)$. We now have the main ingredients necessary for defining our measurement algorithm.

Definition 2.5.

$$\text{In}_\alpha(X) := \left\{ \sum_{i=0}^n |C_i| \mid \cup C_i \subseteq X, \text{ and } \mathbf{int}(C_p) \cap \mathbf{int}(C_q) = \emptyset \text{ for } p \neq q, \text{ and } |C_i| \in M(\alpha) \right\}$$

Where each $C_i \in \mathcal{M}$. $\text{In}_\alpha(X)$ Is called the internal set of X . Note that $\text{In}_\alpha(X)$ can never contain 0, because for each $C_i \subseteq X$, $|C_i| > 0$ (since we only consider intervals with $a < b$).

Definition 2.6.

$$\text{Out}_\alpha(X) := \left\{ \sum_{i=0}^n |C_i| \mid \cup C_i \supseteq X, \text{ and } \mathbf{int}(C_p) \cap \mathbf{int}(C_q) = \emptyset \text{ for } p \neq q, \text{ and } |C_i| \in M(\alpha) \right\}$$

Is called the external or outer set of X .

In both definitions n is a natural number. It should be noted that because n is finite the sums are defined and so each set exists. We make no appeals to infinite sums of surreal numbers. $\text{In}_\alpha(X)$ is to be thought of as the set of all possible measurements which are strictly smaller than or equal to X , the lower bounds of the value of X 's size. $\text{Out}_\alpha(X)$ is to be thought of as the set of values larger than the size of X . We then define the α -estimate of X , denoted $[X]_\alpha$ as follows:

Definition 2.7. The α -Estimate of X , denoted $[X]_\alpha$, is defined

$$[X]_\alpha := \begin{cases} \{ \text{In}_\alpha(X) \mid \text{Out}_\alpha(X) \}, & \text{if } \text{In}_\alpha(X) < \text{Out}_\alpha(X), \\ x & \text{if } x \in \text{In}_\alpha(X) \cap \text{Out}_\alpha(X). \end{cases}$$

This is to be thought of as the most natural size of X which can be assigned on day α , where $M(\alpha)$ represents the lengths of the rulers with which we attempt to capture the size. As the days go by, our instruments become finer and finer, and so in theory we ought to approach the correct size. Perhaps we must wait some finite number of days which exceed the universe's shelf-life, perhaps we must wait some ordinal number of days, or perhaps even that is not enough time to discern the true measure of the set X by our means. In the meantime, we have some theorems and definitions to work upon.

Theorem 2.1. $[]_\alpha$ is well defined.

Proof. Let $i \in \text{In}_\alpha(X)$ and $o \in \text{Out}_\alpha(X)$. We would like to show that $i \leq o$. We will proceed by a series of lemmas. First note that:

$$i = \sum_{j=1}^n |I_j| \text{ and } o = \sum_{k=1}^m |O_k|$$

Where the $\text{int}(I)$'s are pairwise disjoint and the $\text{int}(O)$'s are pairwise disjoint. It suffices to show that if $I_1, \dots, I_n$ and $O_1, \dots, O_m$ are closed intervals such that

$$\cup I_i \subseteq X \subseteq \cup O_j$$

and $\text{int}(I_i) \cap \text{int}(I_j) = \emptyset$ and $\text{int}(O_i) \cap \text{int}(O_j) = \emptyset$ for $i \neq j$, then

$$\sum_{j=1}^n |I_j| \leq \sum_{k=1}^m |O_k|$$

Write $I_i = [a_i, b_i]$ and $O_j = [c_j, d_j]$. Consider the set of obtained by collecting the endpoints

$$E = \{a_1, b_1, \dots, a_n, b_n, c_1, d_1, \dots, c_m, d_m\}$$

and enumerate in increasing order

$$E = \{e_0 < e_1 < e_2, \dots < e_T\}$$



Denote each piece $P_t = [e_{t-1}, e_t]$. Then, if $K = [u, v]$ is any of the $I_1, \dots, I_n, O_1, \dots, O_m$, we can write $K = [e_p, e_q]$. Finally, define

$$\text{indices}(K) = \{ t \mid P_t \subseteq K \}$$

Lemma 2.2. $\text{indices}([e_p, e_q]) = \{p+1, \dots, q\}$

Proof. It is clear to see that, for $P_t \subseteq K$ iff $[e_{t-1}, e_t] \subseteq [e_p, e_q]$ iff $e_p \leq e_{t-1} < e_t \leq e_q$, since our enumeration of E was increasing, this is true iff $p \leq t-1 < t \leq q$ so that $p < t \leq q$ as desired. $\square$

Lemma 2.3. If $K = [u, v] = [e_p, e_q]$ is any of the $I_1, \dots, I_n, O_1, \dots, O_m$ then

$$|K| = \sum_{t \in \text{indices}(K)} |P_t|$$

Proof. By the above lemma,

$$\sum_{t \in \text{indices}(K)} |P_t| = \sum_{t=p+1}^q (e_t - e_{t-1}) = e_q - e_p = v - u = |K|$$

$\square$

Define $\text{BLUE}_i = \text{indices}(I_i)$ and $\text{RED}_i = \text{indices}(O_i)$.

Lemma 2.4. Each of the BLUE_i 's are pairwise disjoint, and similarly for the RED_i 's.

Proof. Suppose $z \in BLUE_i \cap BLUE_j$ with $i \neq j$, this then implies $P_z \subseteq I_i \implies a_i \leq e_{z-1} < e_z \leq b_i$ and $a_j \leq e_{z-1} < e_z \leq b_j$, thus there exists some $x \in (e_{z-1}, e_z)$ such that $x \in \text{int}(I_i)$ and $x \in \text{int}(I_j)$, but these are mutually disjoint. Therefore there must be no such z . $\square$

Now, let $BLUE = \coprod BLUE_i$ and $RED = \coprod RED_i$, then by application of Lemma 2.5 and 2.6 on each set of intervals we attain:

$$\begin{aligned} i &= \sum_{i=1}^n |I_i| = \sum_{t \in BLUE} |P_t| \\ o &= \sum_{j=1}^m |O_j| = \sum_{t \in RED} |P_t| \end{aligned}$$

Lemma 2.5. $BLUE \subseteq RED$.

Proof. Let $t \in BLUE$. So $P_t \subseteq I_k$ for some k . $P_t \subseteq \cup I_i \subseteq \cup O_j$. Pick some $x \in \text{int}(P_t) = (e_{t-1}, e_t) \subseteq P_t \subseteq \cup O_j$, then there exists some O_l such that $x \in O_l$. Write $O_l = [c_l, d_l]$. We claim that $e_{t-1} \geq c_l$, for if $e_{t-1} < c_l$ then $e_t < c_l \leq x < e_t$, but c_l is E and is some e_p , but we cannot have $e_{t-1} < e_p < e_t$. So $c_l \leq e_{t-1}$. By a symmetric argument, $d_l < e_t \implies e_{t-1} < x \leq d_l < e_t$ which leads to a similar contradiction, thus $e_t \leq d_l$. Thus we have proven that $P_t = [e_{t-1}, e_t] \subseteq [c_l, d_l] = O_l$, so $t \in RED_l \subseteq RED$. $\square$

Finally,

$$\sum_{t \in RED} |P_t| - \sum_{t \in BLUE} |P_t| = \sum_{t \in RED - BLUE} |P_t| \geq 0$$

Which implies

$$\sum_{j=1}^m |O_j| - \sum_{i=1}^n |I_i| \geq 0$$

And thus:

$$o \geq i$$

Since i cannot be greater than o we are at least sure that our computation will yield a number whenever $i \neq o$. But what about the case when $i = o$? That is settled in the next lemma:

Lemma 2.6. *If $\exists x, y \in \text{In}_\alpha(X) \cap \text{Out}_\alpha(X)$ then $x = y$.*

Proof. Let $x, y \in \text{In}_\alpha(X) \cap \text{Out}_\alpha(X)$. Per the result above we have $x \leq y \leq x \implies x = y$. □

Now we see that if we find ourselves in the second case of the definition of $[X]_\alpha$ we are guaranteed a unique answer. Both cases being settled we conclude that $[X]_\alpha$ is well-defined. □

2.3 Some Basic Results

Lemma 2.7. *If we can perform a measurement on day α we can perform that same measurement on day $\beta \geq \alpha$, that is, $\text{In}_\alpha(X) \subseteq \text{In}_\beta(X)$ and $\text{Out}_\alpha(X) \subseteq \text{Out}_\beta(X)$.*

Proof. Let $i \in \text{In}_\alpha(X)$, i.e.,

$$i = \sum_{k=0}^n |I_k|$$

with each $I_k \in \mathcal{M}$ and $|I_k| \in M(\alpha)$. Since $M(\alpha) \subseteq M(\beta)$ for $\alpha \leq \beta$ we have $|I_k| \in M(\beta)$ for all k , so $i \in \text{In}_\beta(X)$. A similar argument will hold for $\text{Out}_\alpha(X) \subseteq \text{Out}_\beta(X)$. □

Definition 2.8. If $[X]_\alpha = x \in \text{In}_\alpha(X) \cap \text{Out}_\alpha(X)$ we say that x is the *final measurement* of X . If α is the least ordinal which achieves the final measurement we say that X is *measurable on day α* .

The following lemma justifies this naming convention:

Lemma 2.8. *If $[X]_\alpha = x$ and x is measurable on day α then for all $\beta \geq \alpha$ we have $[X]_\beta = x$.*

Proof. By lemma 2.7 if $x \in \text{In}_\alpha(X)$ and $x \in \text{Out}_\alpha(X)$ then $x \in \text{In}_\beta(X)$ and $x \in \text{Out}_\beta(X)$, therefore $[X]_\beta = x$ by definition. □

Lemma 2.9. $\forall \alpha \in \mathbf{On.}, \alpha > 0 \implies [\emptyset]_\alpha = 0$.

Proof. $\text{In}_\alpha(\emptyset) = \emptyset$, and since $x > 0 \forall x \in \text{Out}_\alpha(\emptyset)$ we see that

$$[\emptyset]_\alpha = \{ \emptyset \mid \dots \} = 0$$

□

Lemma 2.10. $[X]_\alpha \geq 0$.

Proof. $\text{In}_\alpha(X)$ can either be $\emptyset$, in which case $[X]_\alpha$ evaluates to 0, or else if $\text{In}_\alpha(X) \neq \emptyset$, since for every $I \in \mathcal{M}$ $|I| > 0$, $0 < [X]_\alpha$. □

Lemma 2.11. *The sandwich lemma for $[\]_\alpha$. If $i \in \text{In}_\alpha(X)$ and $o \in \text{Out}_\alpha(X)$ then $i \leq [X]_\alpha \leq o$.*

Proof. This holds regardless of whichever clause of the definition is applied. If $[X]_\alpha = \{\text{In}_\alpha(X) \mid \text{Out}_\alpha(X)\}$ then clearly $i < [X]_\alpha < o$ by theorem 2.1. If $[X]_\alpha$ is the final measurement of X with value x then $i \not\leq x$ because $i \in \text{In}_\alpha(X)$. Similarly, $o \not\leq x$. □

Theorem 2.12. $X \subseteq Y$ implies $[X]_\alpha \leq [Y]_\alpha$.

Proof. We proceed by a series of lemmas.

Lemma 2.13. *If $X \subseteq Y$ then $\text{In}_\alpha(X) \subseteq \text{In}_\alpha(Y)$*

Proof. If $i_X \in \text{In}_\alpha(X)$ then $i_X = \sum |C_i|$ such that $\cup C_i \subseteq X \subseteq Y$ so $i_X \in \text{In}_\alpha(Y)$. □

Lemma 2.14. *If $X \subseteq Y$ then $\text{Out}_\alpha(Y) \subseteq \text{Out}_\alpha(X)$.*

Proof. If $o_Y \in \text{Out}_\alpha(Y)$ then $o_Y = \sum |C_i|$ such that $\cup C_i \supseteq Y \supseteq X$, so $o_Y \in \text{Out}_\alpha(X)$. □

Finally, set $x = [X]_\alpha$ and $y = [Y]_\alpha$. There are four cases. If x and y are both final measurements then it follows that $x = y$. If x is a final measurement then $x \in \text{In}_\alpha(Y)$ so $x < y$, if y is a final measurement then $y \in \text{Out}_\alpha(X)$ so $x < y$ and if neither x or y are final then for any $i_X \in \text{In}_\alpha(X) \subseteq \text{In}_\alpha(Y) \implies i_X < y$ and $o_Y \in \text{Out}_\alpha(Y) \subseteq \text{Out}_\alpha(X) \implies o_Y > x$, so by definition of $\leq$ on surreals, we have $x \leq y$. □

Lemma 2.15. *For every closed interval $[a, b] \in \mathcal{M}$ there exists some α such that $[a, b]$ is measurable on day α and $[[a, b]]_\alpha = b - a$.*

Proof. Take

$$\alpha := \beta(b - a)$$

then $[[a, b]] \in M(\alpha)$ and $b - a$ appears in both $\text{In}_\alpha([a, b])$ and $\text{Out}_\alpha([a, b])$. □

It follows that every closed interval with real length is measurable on some finite day or at most on day ω , because every real is born at some natural number stage or on day ω [6].

Note that it is generally not the case that if $X_1 \cap X_2 = \emptyset$ then $[X_1 \cup X_2]_\alpha = [X_1]_\alpha + [X_2]_\alpha$, taking $X_1 = (0, 1)$ and $X_2 = \{0, 1\}$ and $\alpha = 1$, then we see that

$$[(0, 1)]_1 = \{ \mid 1, 2, 3, \dots \} = 0 \text{ and } [\{0, 1\}]_1 = \{ \mid \} = 0$$

However,

$$[X_1 \cup X_2]_1 = [[0, 1]]_1 = 1 \neq [X_1]_1 + [X_2]_1 = 0$$

Lemma 2.16. *The translation lemma for $[\]_\alpha$. If $[X]_\alpha = x$ then $[X + k]_\alpha = x$ where $k \in \mathbf{No.}$ and $X+k = \{ x + k \mid x \in X \}$*

Proof. If $i \in \text{In}_\alpha(X)$ then

$$i = \sum_{j=1}^n |I_j| \text{ and } \bigcup I_j \subseteq X$$

Where each $I_j = [a_j, b_j]$. If $[a_j, b_j] \subseteq X$ then $I'_j = [a_j + k, b_j + k] \subseteq X + k$ and since $|I'_j| = |I_j|$ we have

$$i = \sum_{j=1}^n |I'_j| \text{ and } \bigcup I'_j \subseteq X,$$

so $i \in \text{In}_\alpha(X+k)$. Note that $X = (X + k) - k$ so any $i' \in \text{In}_\alpha(X+k)$ will appear in $\text{In}_\alpha(X)$. A similar proof will show that $\text{Out}_\alpha(X) = \text{Out}_\alpha(X+k)$. $\square$

3 Some Computations with the α -Estimate

We begin this section with one further definition.

Definition 3.1.

$$\text{Inf}(\alpha) = \min \{ x \mid x \in M(\alpha), x > 0 \}$$

For all $\alpha \in \mathbf{On.}$, $\alpha \neq 0$.

This is to be thought of as "the smallest value measurable on day α ." Clearly there is no such value on day 0. On day ω we receive our first infinitesimal ruler of length ϵ , but we cannot use this ruler to measure arbitrary subsets of $\mathbb{R}$. This is because we can only string together finitely many rulers when we conduct our internal and outer measurements, so even a simple interval such as $[0, 2]$ is far too large for our ϵ -ruler to handle on its own.

3.1 An Illustrative Illustrated Example

Let $Q = [2, 3] \cup [3\frac{1}{3}, 3\frac{2}{3}]$ whose expected length is $1\frac{1}{3}$. We shall evaluate $[Q]_\alpha$ on days $\alpha \in \{1, 2, 3, \omega\}$.

Day 1: We have a ruler of length 1, so our under-estimate is 1 because $[2, 3] \subseteq Q$. We can over-estimate with a values of 2 because $[2, 3] \cup [3, 4] \supseteq Q$. **Fig A** shows what the measurement looks like. Note that the two measuring sticks are placed end to end, this corresponds to the fact that $\mathbf{int}([2, 3]) \cap \mathbf{int}([3, 4]) = \emptyset$. We estimate the size of Q to be $1\frac{1}{2}$.

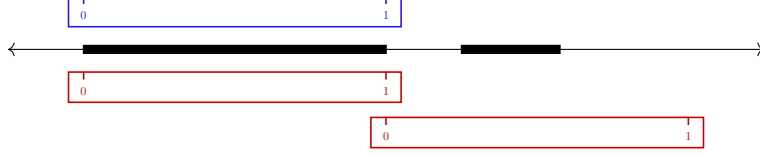


Fig A: Measuring the set Q on day 1.

$$[Q]_1 = \{ 1 \mid 2, 3, 4, \dots \} = 1\frac{1}{2}$$

Day 2: We get new rulers of length 2 and $\frac{1}{2}$. We can treat the ruler of length $\frac{1}{2}$ as a new tic mark on our ruler measuring the same length. At this stage the only beneficial device is the ruler of length $\frac{1}{2}$ because it affords us finer measurements.

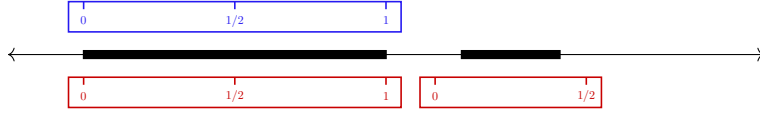


Fig B: Measuring the set Q on day 2.

$$[Q]_2 = \{ \frac{1}{2}, 1 \mid 1\frac{1}{2}, 2, 2\frac{1}{2}, \dots \} = 1\frac{1}{4}$$

Day 3: We get new rulers of length 3 and $\frac{1}{4}$, but the ruler of length $\frac{1}{4}$ is really all that's useful to us. Note that we do not have a ruler of length $\frac{1}{3}$ and will not until day ω .

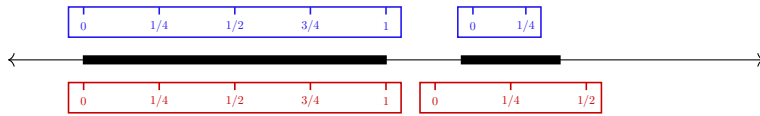


Fig C: Measuring the set Q on day 3.

$$[Q]_3 = \{ \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1, 1\frac{1}{4} \mid 1\frac{1}{2}, \dots \} = 1\frac{3}{8}$$

Day ω : We get many new rulers, one for each real number, in particular a ruler of length $\frac{1}{3}$. Thus our lower-bound for our measurement is $1\frac{1}{3}$ and so is our upper-bound. Therefore we conclude the final size of Q is $1\frac{1}{3}$. Note that we also get rulers of length ω and length ϵ . The former is clearly no use to us, but so is the latter, for we cannot simply string together finitely many ϵ -length rulers, for our lower estimates would be $\epsilon, 2\epsilon, 3\epsilon, \dots$ and we could not find an upper estimate

if we were to use infinitesimal length rulers exclusively.



Fig D: Measuring the set Q on day ω .
 $[Q]_\omega = 1\frac{1}{3}$

3.2 Computations

We perform some basic computations with the α -Estimate, taking $\alpha \in \mathbb{N}$ and our set $X \in \{(0, 1), [0, 1]\}$.

X	α	$\text{In}_\alpha(X)$	$\text{Out}_\alpha(X)$	$[X]_\alpha$
$(0, 1)$	0	$\emptyset$	$\emptyset$	0
$[0, 1]$	0	$\emptyset$	$\emptyset$	0
$(0, 1)$	1	$\emptyset$	$\{1, 2, \dots\}$	0
$[0, 1]$	1	$\{1\}$	$\{1, 2, \dots\}$	1
$(0, 1)$	2	$\{\frac{1}{2}\}$	$\{1, 1\frac{1}{2}, \dots\}$	$\frac{3}{4}$
$[0, 1]$	2	$\{\frac{1}{2}, 1\}$	$\{2, 2\frac{1}{2}, \dots\}$	$1\frac{1}{2}$
$(0, 1)$	3	$\{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$	$\{1\frac{1}{4}, \dots\}$	$\frac{7}{8}$
$[0, 1]$	3	$\{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$	$\{1, 1\frac{1}{4}, \dots\}$	1
$(0, 1)$	4	$\{\dots, \frac{7}{8}\}$	$\{1, 1\frac{1}{8}, \dots\}$	$\frac{15}{16}$
$[0, 1]$	4	$\{\dots, 1\}$	$\{1, \frac{1}{4}, 1\frac{3}{8}, \dots\}$	1

What about for transfinite values of α ?

X	α	$\text{In}_\alpha(X)$	$\text{Out}_\alpha(X)$	$[X]_\alpha$
$(0, 1)$	ω	$\{\dots, 1 - \epsilon\}$	$\{1, 1 + \epsilon, \dots\}$	$1 - \frac{\epsilon}{2}$
$[0, 1]$	ω	$\{\dots, 1 - \epsilon, 1\}$	$\{1, 1 + \epsilon, \dots\}$	1
$(0, 1)$	$\omega + 1$	$\{\dots, 1 - \frac{\epsilon}{2}\}$	$\{1, 1 + \frac{\epsilon}{2}, \dots\}$	$1 - \frac{\epsilon}{4}$
$[0, 1]$	$\omega + 1$	$\{\dots, 1 - \frac{\epsilon}{2}, 1\}$	$\{1, + \frac{\epsilon}{2}, \dots\}$	1

This would suggest for transfinite $1 < \alpha \in \mathbf{On.}$, $[(0, 1)]_\alpha = 1 - \frac{\inf(\alpha)}{2}$ and motivates the following lemma:

Lemma 3.1. *For all open intervals (a, b) and all values $\epsilon > 0$ there exists an α such that $\forall \kappa \geq \alpha$ we have*

$$|[(a, b)]_\kappa - (b - a)| < \epsilon$$

Proof. Set

$$\alpha := \max\{\beta(b-a), \beta(b-a-\epsilon)\}$$

So $[a, b] \in \text{Out}_\alpha((a, b))$ and $[a + \frac{\epsilon}{2}, b - \frac{\epsilon}{2}] \in \text{In}_\alpha((a, b))$ and thus

$$b - a - \epsilon < [(a, b)]_\alpha < b - a$$

As desired. By lemma 2.7 we can conclude that for all $\kappa \geq \alpha$

$$|[(a, b)]_\kappa - (b - a)| < \epsilon$$

□

Definition 3.2. We say that a set X is α -null if and only if $[X]_\alpha < \inf(\alpha)$.

Definition 3.3. We define a relation between two sets, $=_\alpha$, and say $X =_\alpha Y$ if and only if $|[X]_\alpha - [Y]_\alpha| < \inf(\alpha)$.

Lemma 3.2. X is α -null if and only if $X =_\alpha \emptyset$.

Proof. Trivial. □

3.3 Sometimes ∞ is All You Need

Consider the sets

$$R = \bigcup_{n \in \mathbb{N}} [n, n+1]$$

$$B = R \cup [\omega, \omega+1]$$

Evaluating the α -estimate of R and B on any finite day n yields:

$$[R]_n = \{1, 2, 3, \dots \mid \} = \omega = [B]_n$$

On day ω we may use the interval of length ω , $[0, \omega]$ to contain R , but we cannot use any interval of length ω to contain all of B as either one of the finite natural number intervals will be excluded, or else the last interval $[\omega, \omega+1]$ will be. Thus, on day ω these two sets have distinct measure:

$$[R]_\omega = \{1, 2, 3, \dots \mid \omega, \dots \} = \omega - 1$$

$$[B]_\omega = \{1, 2, 3, \dots \mid \omega + 1, \dots \} = \omega$$

What happens on day $\omega + 1$? On this day, $\omega - 1$ is born and so we may use the interval $[0, \omega - 1]$ to cover R , and so this appears in $\text{Out}_{\omega+1}(R)$. Similarly, we can use that same interval to cover the intervals with natural number endpoints in B , then use one interval of length 1 to cover the remainder. So on day $\omega + 1$ we have:

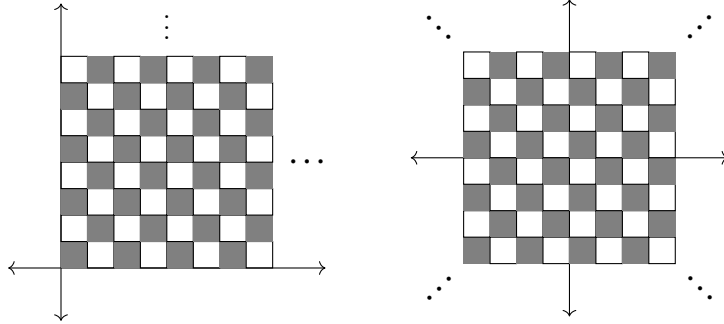


Fig. A shows a "chessboard" with each square of unit area, the left corner at the origin, figure **Fig. B** shows a chessboard which goes on infinitely in all four cardinal directions. How might the areas of their shaded regions be calculated?

$$[R]_{\omega+1} = \{1, 2, 3, \dots \mid \omega - 1, \dots\} = \omega - 2$$

$$[B]_{\omega+1} = \{1, 2, 3, \dots \mid \omega, \dots\} = \omega - 1$$

Indeed, let χ be any surreal number greater than all natural numbers, then take $\Gamma = \beta(\chi) + 1$, then $\chi \in \text{Out}_\Gamma(R)$ and $\chi \in \text{Out}_\Gamma(B)$. This result makes the following statement precise:

$$\lim_{\alpha \rightarrow \Omega} [R]_\alpha = \infty = [B]_\alpha \lim_{\Omega \leftarrow \alpha}$$

This phenomenon is quite striking. When introducing non-standard number systems to try to evaluate infinite quantities we do so because we feel that we have not adequately distinguished between them, that there should be some discernible difference between the two. This calculation would seem to suggest that, on the contrary, ∞ is the "most accurate" answer that one can give, but nonetheless at transfinite resolutions these two sets would appear to have distinct sizes. It is important to note that intervals like $[0, \omega]$ will always evaluate to ω , however, and the interval $[0, \omega + 1]$ reaches its final measurement of $\omega + 1$ on day $\omega + 1$ per lemma 2.15.

4 Generalization to Area and Volume

Let $C \in \mathcal{M}^n$, that is C is to be thought of as an n -dimensional rectangle $C = \prod_{i=1}^n I_i$. By convention we say the k th element of the product is C^k . For $X \subseteq \mathbf{No}^n$ define the following:

Definition 4.1. For $C \in \mathcal{M}^n$ define $\mathbf{vol}(C) = \prod_{i=1}^n |I_i|$.

Definition 4.2.

$$\text{In}_\alpha^n(X) := \left\{ \sum_{i=0}^n \text{vol}(C_i) \mid \cup C_i \subseteq X, \text{ and } \text{int}(C_p) \cap \text{int}(C_q) = \emptyset \text{ for } p \neq q, \text{ and } |C_i^k| \in M(\alpha) \right\}$$

Definition 4.3.

$$\text{Out}_\alpha^n(X) := \left\{ \sum_{i=0}^n \text{vol}(C_i) \mid \cup C_i \supseteq X, \text{ and } \text{int}(C_p) \cap \text{int}(C_q) = \emptyset \text{ for } p \neq q, \text{ and } |C_i^k| \in M(\alpha) \right\}$$

Definition 4.4. The n-dimensional α -Estimate of X, denoted $[X]_\alpha^n$, is defined

$$[X]_\alpha^n := \begin{cases} \{ \text{In}_\alpha^n(X) \mid \text{Out}_\alpha^n(X) \}, & \text{if } \text{In}_\alpha^n(X) < \text{Out}_\alpha^n(X), \\ x & \text{if } x \in \text{In}_\alpha^n(X) \cap \text{Out}_\alpha^n(X). \end{cases}$$

It can be shown that these definitions satisfy all of the properties one could reasonably expect, the proofs in section 2 can be modified slightly to yield similar results.

When considering the question of say, the areas of the shaded regions in **fig (A)** and **fig (B)**, one is burdened by which considerations to choose. It might be reasonable to say that A and B have areas in proportion A:B = 1 : 4. Perhaps they should be equal because the shaded region of A can be put into one-to-one correspondence with B. Our definition allows us to skip the philosophizing and we can happily "shut up and calculate" to get our answer. On some finite day we have $[A]_n^2 = [B]_n^2 = \{1, 2, 3, \dots \mid \} = \omega$. Similarly on day ω we can cover A with the rectangle $[0, \omega] \times [0, \omega]$ and B with the rectangle $[\frac{-\omega}{2}, \frac{\omega}{2}] \times [\frac{-\omega}{2}, \frac{\omega}{2}]$ of area ω^2 , so on day ω we get

$$[A]_\omega^2 = [B]_\omega^2 = \{1, 2, 3, \dots \mid \omega^2, \dots\} = \omega$$

If we wait for the day on which $\sqrt{\omega}$ is born, however, we can use the coverings $[0, \sqrt{\omega}] \times [0, \sqrt{\omega}]$ and $[\frac{-\sqrt{\omega}}{2}, \frac{\sqrt{\omega}}{2}] \times [\frac{-\sqrt{\omega}}{2}, \frac{\sqrt{\omega}}{2}]$ of area ω for the sets A and B respectively. In general if χ is any infinite surreal number it will appear in the outer set on day at most $\beta(\sqrt{\chi})$ of both sets, and so again we find the limit as $\alpha \rightarrow \Omega$ goes to ∞ for both $[A]_\alpha^2$ and $[B]_\alpha^2$.

5 Error with respect to Lebesgue-Measure

The natural question arises, if X is measurable, what can be said about the quantity:

$$|\mu(X) - [X]_\alpha|$$

This question is more complicated than it would seem. For instance, if $X \subseteq \mathbb{R}$ then, in general

$$[X]_\alpha = 0$$

This is because there are no surreal intervals which are subsets of $\mathbb{R}$, so $\text{In}_\alpha(X) = \emptyset$. Another question arises: suppose that we were to restrict ourselves to only *real-valued* measurements, would our α -Estimates be the same? We shall define exactly this operation, and, since there is a natural correspondence between surreal intervals with real endpoints and real intervals, we will be able capture the length the sizes of certain subsets of the real line.

Definition 5.1. The Real-Valued internal set, denoted $\text{In}_\alpha^\mathbb{R}(X)$ is defined:

$$\left\{ \sum_{i=0}^n |C_i| \mid \cup C_i \subseteq X, \text{ and } \mathbf{int}(C_p) \cap \mathbf{int}(C_q) = \emptyset \text{ for } p \neq q, \text{ and } |C_i| \in M(\alpha) \right\}$$

Where each $C_i \subseteq \mathbb{R}$ is a closed interval with distinct endpoints. Where each $I \subseteq \mathbb{R}$

$\text{Out}_\alpha^\mathbb{R}(X)$ is defined similarly.

Definition 5.2.

$$\text{Out}_\alpha^\mathbb{R}(X) :=$$

$$\left\{ \sum_{i=0}^n |C_i| \mid \cup C_i \supseteq X, \text{ and } \mathbf{int}(C_p) \cap \mathbf{int}(C_q) = \emptyset \text{ for } p \neq q, \text{ and } |C_i| \in M(\alpha) \right\}$$

Definition 5.3. The real-valued α -Estimate of X , denoted $[X]_\alpha^\mathbb{R}$, is defined

$$[X]_\alpha^\mathbb{R} := \begin{cases} \{ \text{In}_\alpha^\mathbb{R}(X) \mid \text{Out}_\alpha^\mathbb{R}(X) \}, & \text{if } \text{In}_\alpha^\mathbb{R}(X) < \text{Out}_\alpha^\mathbb{R}(X), \\ x & \text{if } x \in \text{In}_\alpha^\mathbb{R}(X) \cap \text{Out}_\alpha^\mathbb{R}(X) \end{cases}$$

Note that there is a natural correspondence between intervals $[a, b] \subseteq \mathbb{R}$ and intervals $(a, b) \in \mathcal{M}$ where a and b are real numbers, and thus a natural correspondence between finite unions of intervals and finite unions of elements of $\mathcal{M}$ with real endpoints.

Theorem 5.1. If $C^\mathbb{R}$ is an interval $[a, b] \subseteq \mathbb{R}$ and $C = [a, b]$ is its corresponding surreal counterpart, then:

$$[C^\mathbb{R}]_\alpha^\mathbb{R} = [C]_\alpha$$

for $\alpha < \omega$.

Proof. First, it is clear that $\text{In}_\alpha^\mathbb{R}(C^\mathbb{R}) \subseteq \text{In}_\alpha^\mathbb{R}(C)$. Let $x \in \text{In}_\alpha(C)$, so that we have

$$C_1 \cup C_2 \cup \dots \cup C_n \subseteq C = [a, b]$$

With $\sum |C_i| = x$ and each C_i pairwise disjoint. We would like to convert these into a sequence $C'_1, C'_2, \dots, C'_n$ where each C'_i has real endpoints, and $|C'_i| = |C_i|$.

Since $\alpha < \omega$ we know that the length of each C_i is some dyadic rational. In particular we can write $C_1 = [a_1, b_1]$ and note that $b_1 - a_1$ is rational. If a_1 and b_1 are real, then we can take $C_1 = C'_1$. Otherwise, we can write

$$[a_1, b_1] = [c_1 \pm \epsilon_1, d_1 \pm \epsilon_1]$$

Where c_1 and d_1 are real numbers and ϵ_1 is some infinitesimal. This is because, writing a_1 in Conway normal form

$$a \leq a_1 = \sum_{\beta \in \alpha} r_\beta * \omega^{y_\beta}$$

Since a_1 is bounded by the real a this means $y_0 \leq 0$, and so each non-zero term is either a real or an infinitesimal. (For details, see [6] page 33, theorem 21).

So we have $a \leq c_1 \pm \epsilon$, and we can set $C'_1 = [c_1, d_1]$. For $C_2 = [a_2, b_2]$, we can similarly write $C_2 = [c_2 \pm \epsilon_2, d_2 \pm \epsilon_2]$. From the above we have:

$$d_1 \pm \epsilon_1 \leq c_2 \pm \epsilon_2$$

We claim that $c_2 \geq d_1$, for if not we have

$$\pm \epsilon_1 \mp \epsilon_2 \leq c_2 - d_2$$

Which we can simplify to

$$\epsilon < R$$

Where ϵ denotes that resultant infinitesimal quantity and R is the real-valued difference of d_1 and c_2 . Now, if R is positive or 0 then we are done and $c_2 \geq d_1$. If R is negative, then ϵ must be negative, and this would imply the existence of an infinitesimal number greater than some real number, which is a contradiction. Now that we have $d_1 \leq c_2$, we can safely take $C'_2 = [c_2, d_2]$.

We can continue this way until we get to $C_n = [a_n, b_n] = [c_n \pm \epsilon_n, d_n \pm \epsilon_n]$, by the reasoning above we know that $d_{n-1} \leq c_n$ and since

$$d_n \pm \epsilon_n \leq b$$

we can conclude that $d_n \leq b$. Thus we can take $C'_n = [c_n, d_n]$. Note that each $[c_i, d_i]^\mathbb{R} \subseteq C^\mathbb{R}$ and that each of the $\mathbf{int}(C'_i)$ are disjoint and that $|C_i| = |C'_i|$. That is,

$$C'_1 \cup C'_2 \cup \dots \cup C'_n \subseteq C^\mathbb{R}$$

and $\sum |C'_i| = x$, and we may conclude that $x \in \text{In}_\alpha^\mathbb{R}(C^\mathbb{R})$. A very similar argument can be carried out to show that

$$\text{Out}_\alpha^\mathbb{R}(C^\mathbb{R}) = \text{Out}_\alpha(C)$$

So we may conclude that

$$[C^\mathbb{R}]_\alpha^\mathbb{R} = [C]_\alpha$$

□

Theorem 5.2. *If C is a closed interval, and $\epsilon^\mathbb{R}$ is a real number then there exists an $\kappa < \omega$ such that $\alpha \geq \kappa$ implies*

$$|\mu(C) - [C]_\alpha^\mathbb{R}| \leq \epsilon^\mathbb{R}$$

Proof. By **Theorem 5.1** We have

$$|\mu(C) - [C]_\alpha^\mathbb{R}| = |\mu(C) - [C]_\alpha|$$

Let δ be a dyadic rational $0 < \delta < \epsilon^\mathbb{R}$. We will find some κ for which

$$|b - a - [C]_\kappa| \leq \delta$$

Choose

$$[d_1, d_2] \subseteq [a, b] \subseteq [D_1, D_2]$$

Such that d_1, d_2, D_1 and D_2 are all dyadic rationals that satisfy

$$d_1 - a \leq \frac{\delta}{4} \quad b - d_2 \leq \frac{\delta}{4}$$

□

Note that this is not generally the case for any measurable set X . Take for example the set $A = (0, 1) - \mathbb{D}$ where $\mathbb{D}$ is the set of dyadic rationals. $\mu(A) = 1$ but $[A]_\alpha^\mathbb{R} = 0$ for finite α . Indeed, $\text{In}_\alpha^\mathbb{R}(A) = \emptyset$, for there can be no interval that "fits" inside of A .

6 On Vitali Sets

Let V be a set obtained via the typical Vitali set construction, it is well known that there are no (real) closed non-degenerate sub-intervals which are contained in V , that is, $\text{In}_\alpha^\mathbb{R}(V) = \text{In}_\alpha(V) = \emptyset$, so we have the result

$$[V]_\alpha^\mathbb{R} = 0$$

7 Future Directions

We have defined the α -Estimate, but there are many questions (low-hanging fruit!) left to explore. In our analysis we restricted ourselves to only finite sums in the inner and outer sets, however, work of Alling [2] shows how we can stratify the Field of surreal numbers into fields with power-series structure $\zeta\mathbf{No.}$, where $\zeta > 0$ is an ordinal number. Could we add an extra parameter ζ to the α -Estimate and take a great number of sums? How will this affect the resulting measurements? We defined the α -Estimate for higher dimensions but did not explore the structure to completion. Does the $n - \alpha$ Estimate of structures of dimension $< n$ tend to 0 as $\alpha \rightarrow \mathbf{On.}$? $[]_\alpha$ has a clear predecessor, which is the Jordan-Peano measure, is there a similar surreal analogue for the Lebesgue measure? One which hopefully avoids the defect of not being able to give the "right" answer to the size of A ? These are just a few fun directions.

References

- [1] Haidar Al-Dhalimy and Charles J. Geyer. Surreal time and ultratasks. *The Review of Symbolic Logic*, 9(4):836–847, 2016.
- [2] Norman L. Alling. Conway’s field of surreal numbers. *Transactions of the American Mathematical Society*, 287(1):365–386, 1985.
- [3] Norman L. Alling. *Foundations of Analysis over Surreal Number Fields*, volume 141 of *North-Holland Mathematics Studies*. North-Holland, 1987.
- [4] Merlin Carl, Tim Fischbach, Peter Koepke, Russell Miller, Miriam Nasfi, and Gregor Weckbecker. The basic theory of infinite time register machines. *Archive for Mathematical Logic*, 49(2):249–273, 2010.
- [5] Eddy Keming Chen and Daniel Rubio. Surreal decisions. *Philosophy and Phenomenological Research*, 100(1), 2020.
- [6] John H. Conway. *On Numbers and Games*. A K Peters, 2nd edition, 1976.
- [7] John H. Conway Elwyn Berlekamp and Richard K. Guy. *Winning Ways for Your Mathematical Plays*. CRC Press, 2nd edition, 2001.
- [8] J. Dmitri Gallow. Surreal probabilities. 2024. Unpublished manuscript.
- [9] Robert Goldblatt. *Non-Standard Analysis*. 1998.
- [10] Joel David Hamkins and Andy Lewis. *Infinite Time Turing Machines*. 1998.
- [11] Donald E. Knuth. *Surreal Numbers*. Addison-Wesley Professional, 1st edition, 1974.
- [12] Jon Pérez Laraudogoitia. A beautiful supertask. *Mind*, 105, 1996.